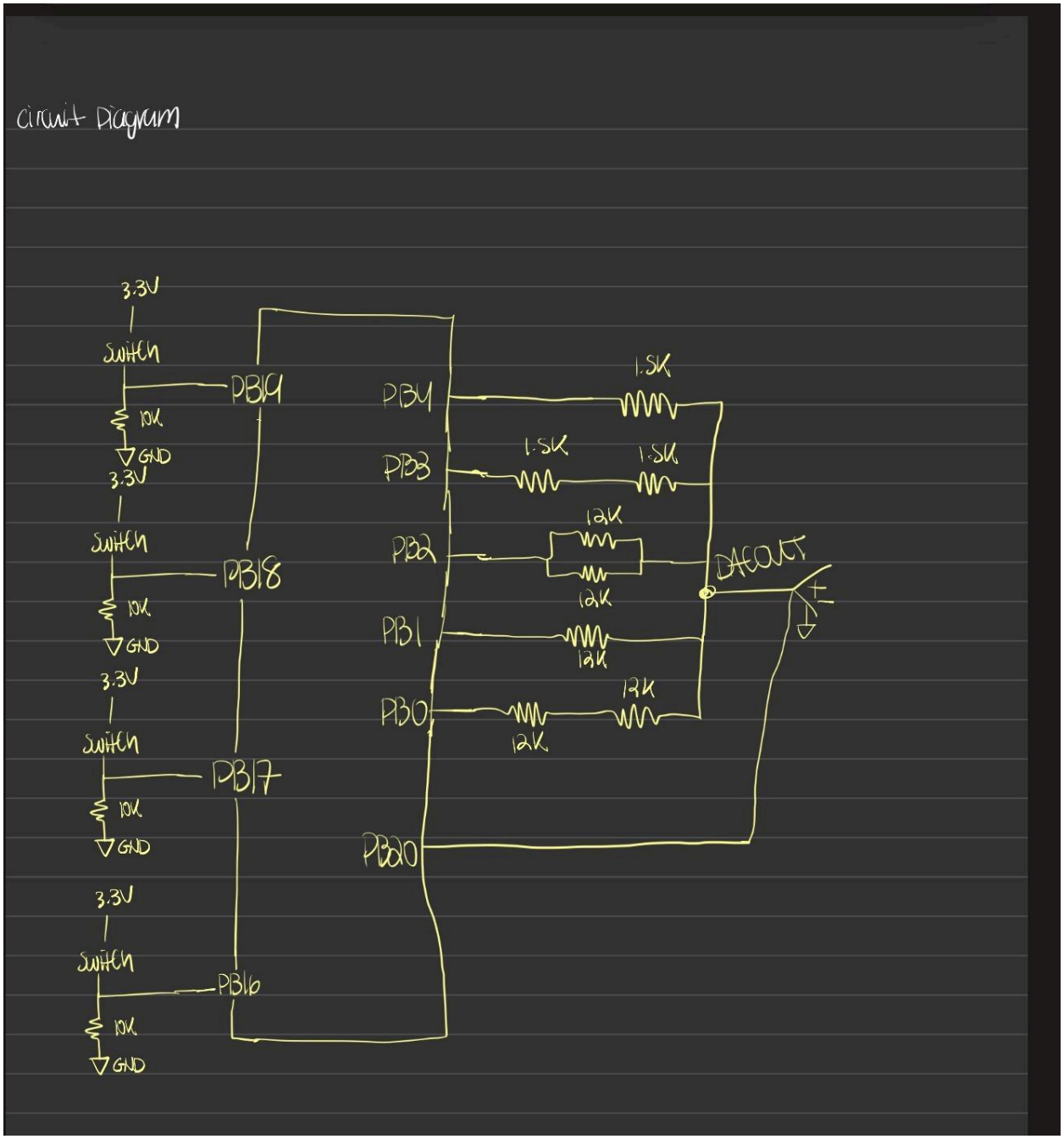
1 Maaz Ahmed, Professor Holt, ma69299

2



3

Software Design

Data Structures:

Sinewave [32]

16	19	22	25	27	29	30	31
31	31	30	29	29	25	22	19
16	13	10	7	7	3	2	1
1	1	2	3	3	7	10	13

32-entry sinewave look-up table used by the systick interrupt to generate audio samples.

uint32_t Index;

[0] → increments

index variable that steps through the sine wave table.

Data Flow Diagram:

[push Buttons]

↓

key_In()

↓

main() (selects period)

↓

sound_start (period)

↓

systick timer (interrupts)

↓

systick_Handler()

↓

sinewave[index]

↓

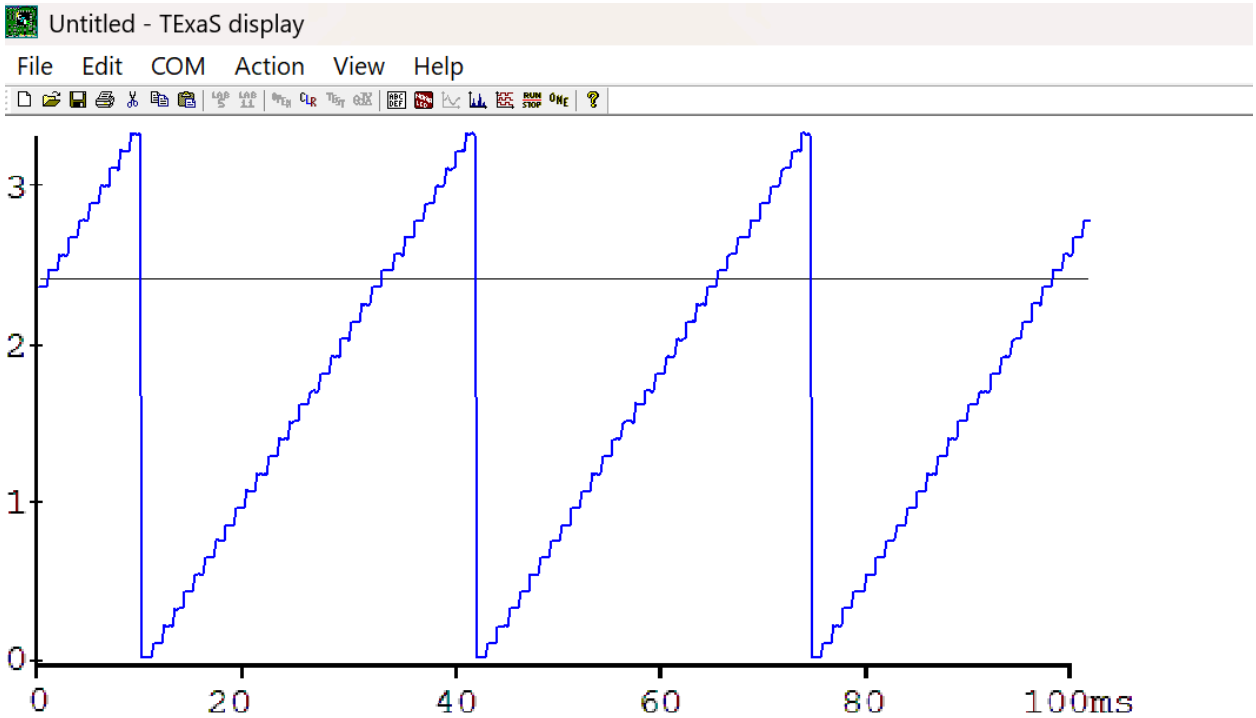
DAC_Out()

↓

DAC Hardware

↓

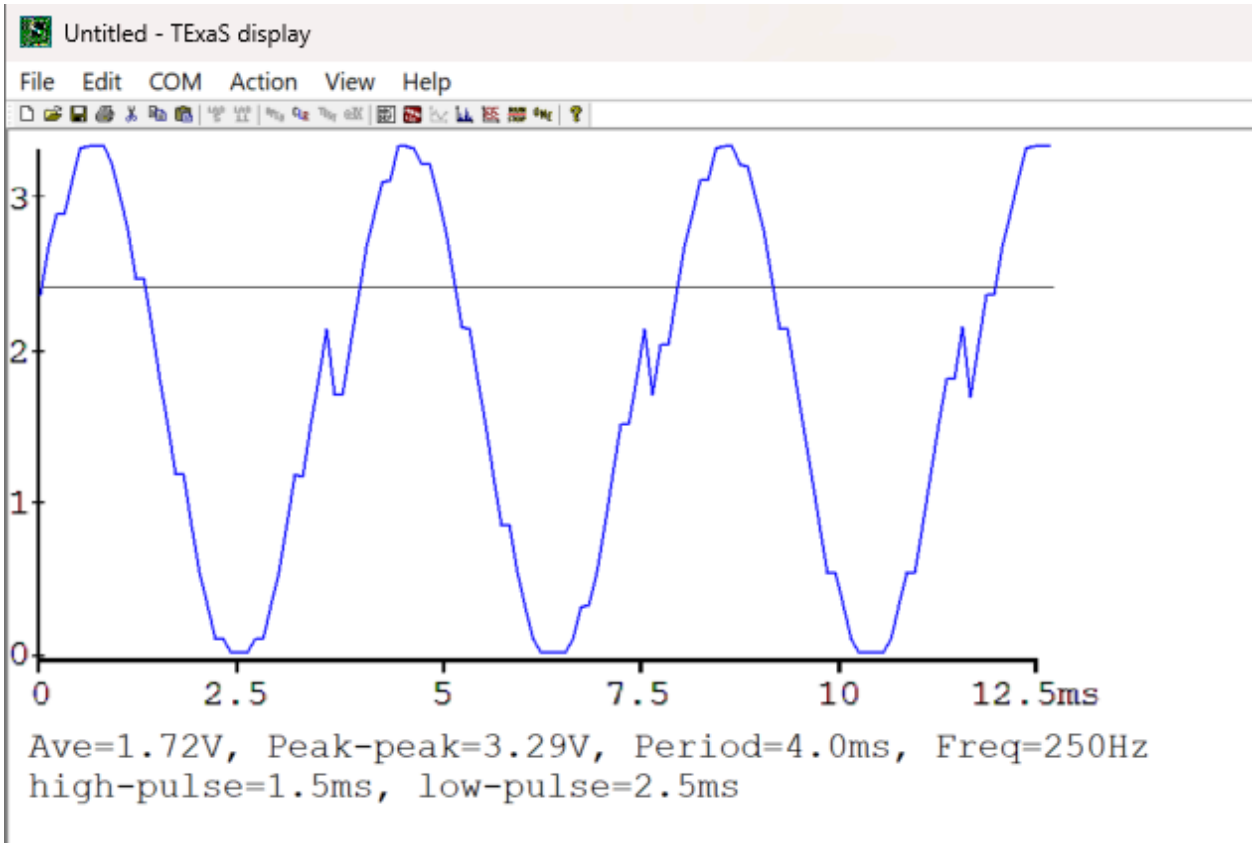
Analog Audio output.



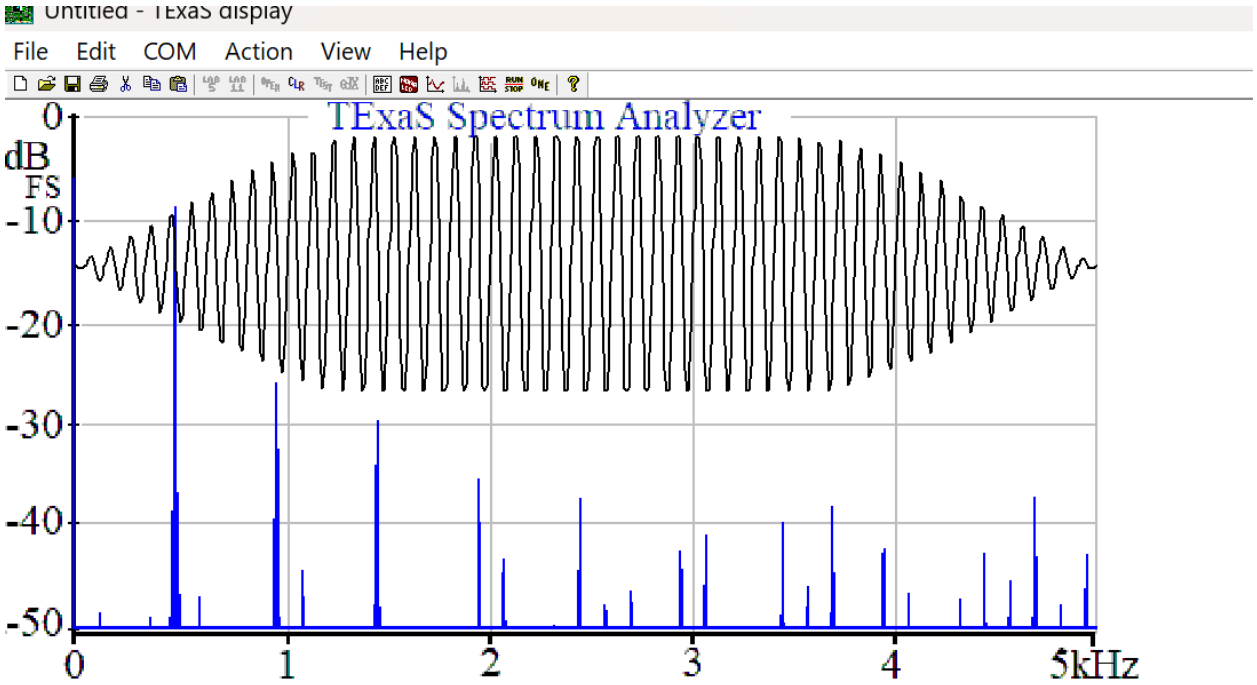
Ave=1.67V, Peak-peak=3.29V, Period=32.6ms, Freq= 31Hz
high-pulse=10.2ms, low-pulse=22.4ms

5

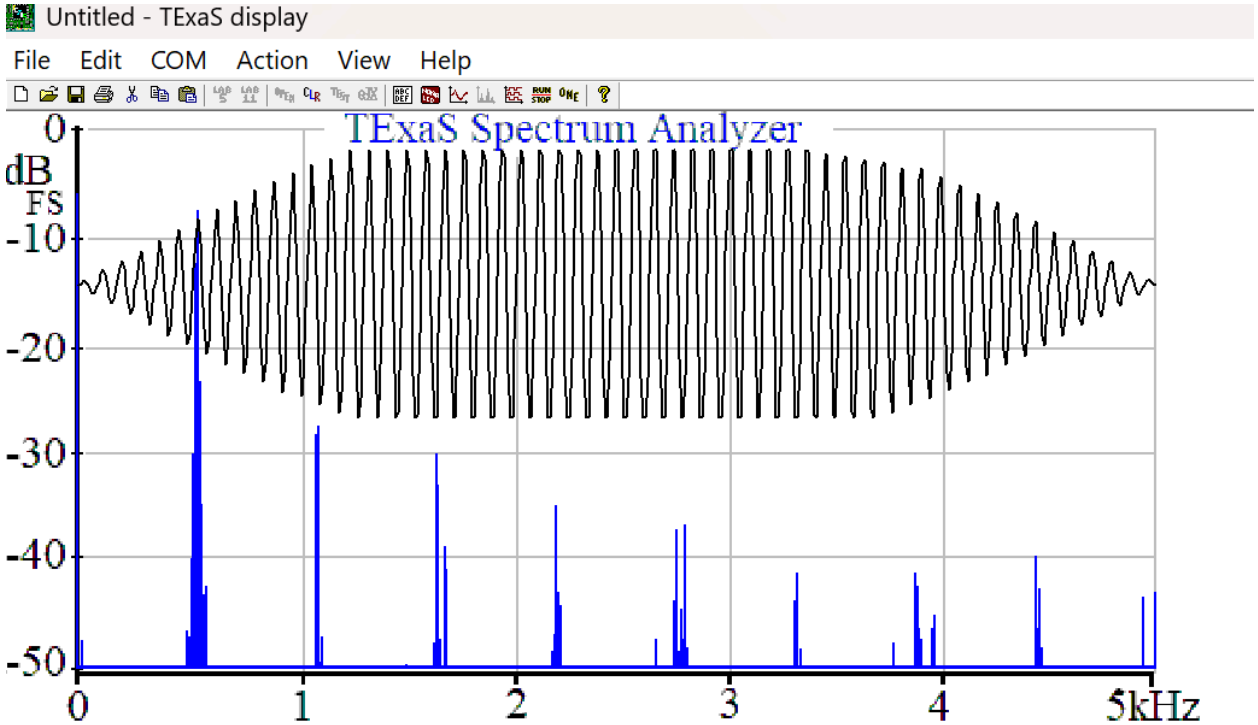
```
Lab 5, Spring 2026, Step 4. Debug switches
EID1= MA69299
Switch= 0x8
Switch= 0x0
Switch= 0x4
Switch= 0x0
Switch= 0x2
Switch= 0x0
Switch= 0x1
Switch= 0x0
```


Untitled - TExaS display
File Edit COM Action View Help
3
2
1
0
0
2.5
5
7.5
10
12.5ms
Ave=1.72V, Peak-peak=3.29V, Period=4.0ms, Freq=250Hz
high-pulse=1.5ms, low-pulse=2.5ms

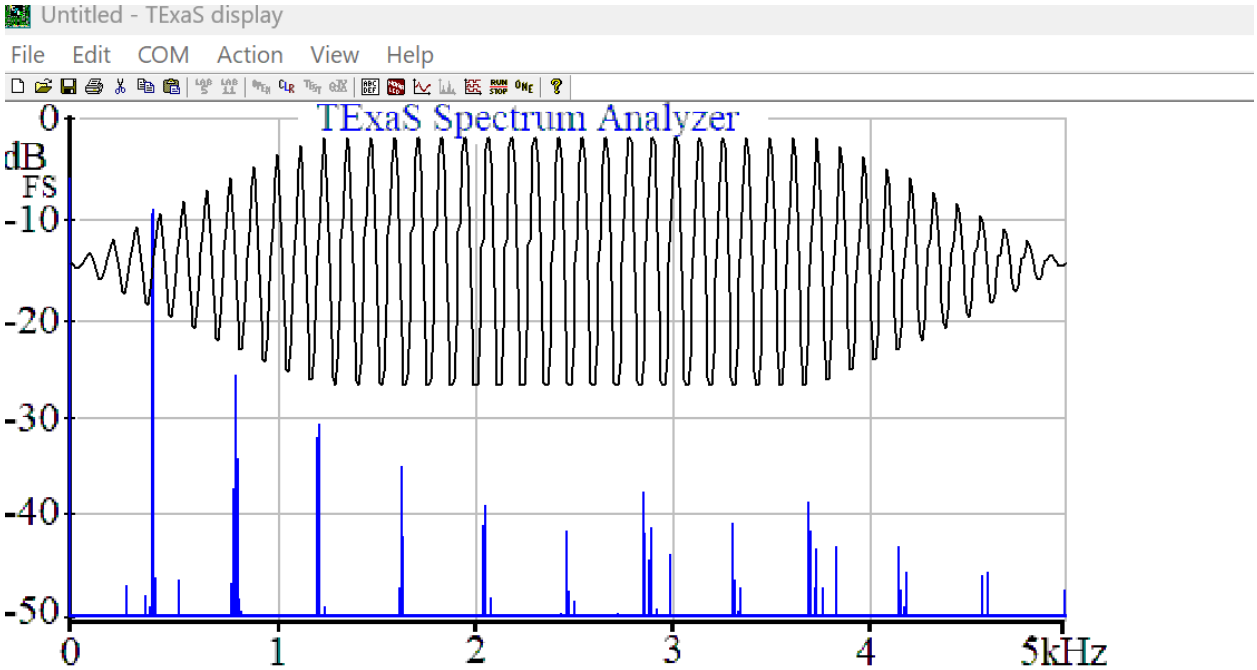
7



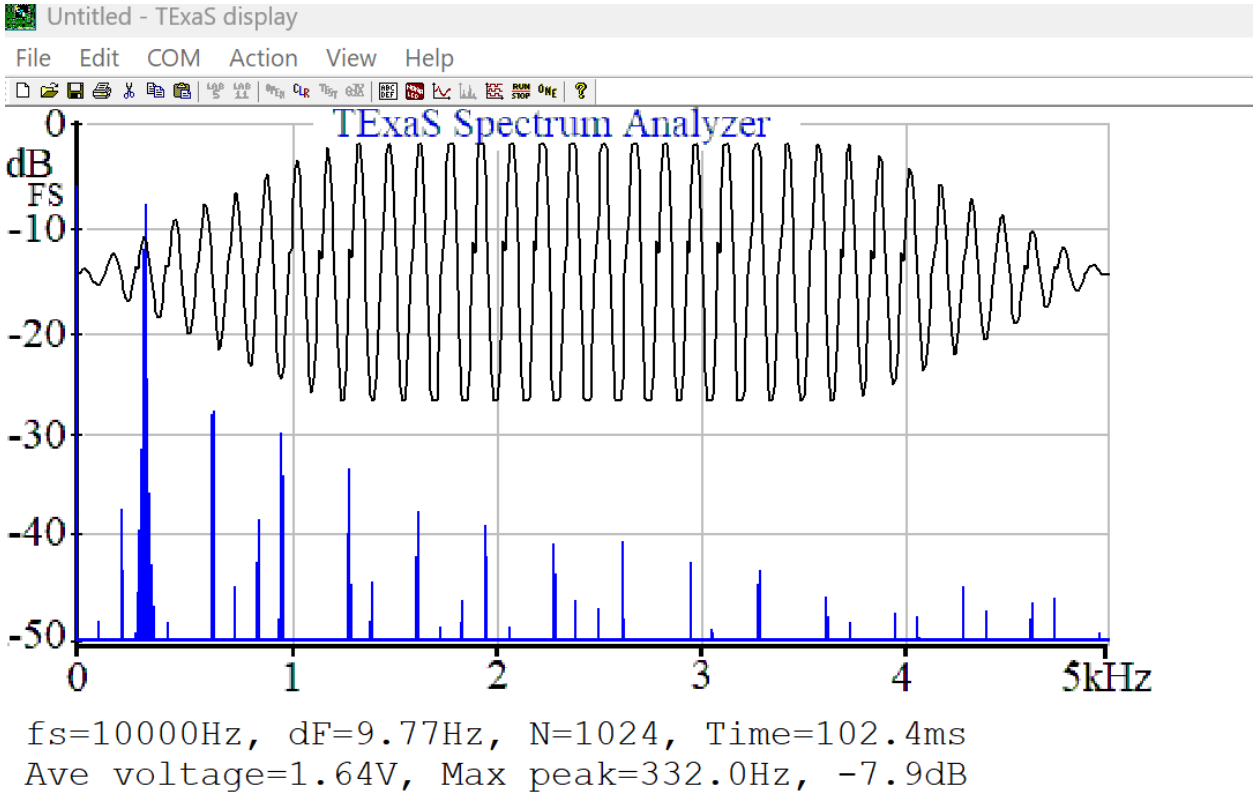
fs=10000Hz, dF=9.77Hz, N=1024, Time=102.4ms
Ave voltage=1.62V, Max peak=498.0Hz, -8.9dB



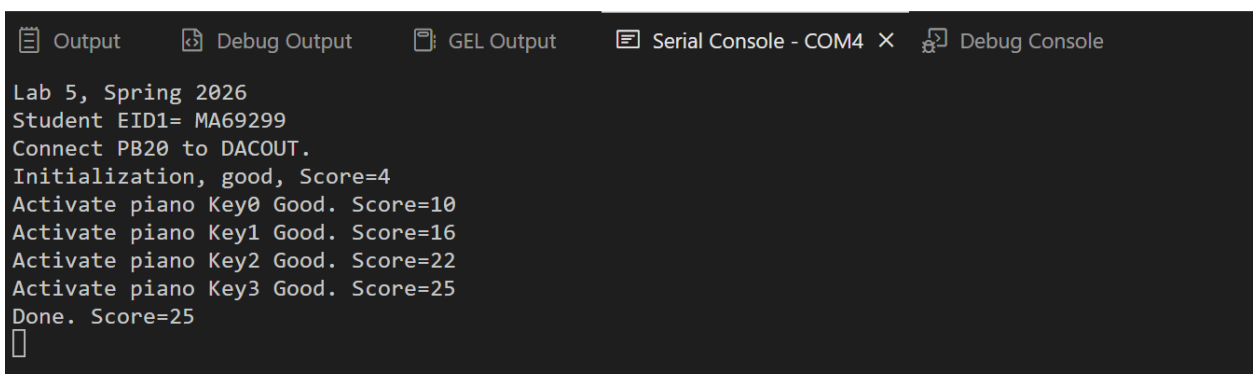
fs=10000Hz, dF=9.77Hz, N=1024, Time=102.4ms
Ave voltage=1.64V, Max peak=556.6Hz, -7.8dB



fs=10000Hz, dF=9.77Hz, N=1024, Time=102.4ms
Ave voltage=1.62V, Max peak=419.9Hz, -9.3dB


Untitled - TExaS display
File Edit COM Action View Help
TExaS Spectrum Analyzer
dB
FS
0
-10
-20
-30
-40
-50
0
1
2
3
4
5kHz
fs=10000Hz, dF=9.77Hz, N=1024, Time=102.4ms
Ave voltage=1.64V, Max peak=332.0Hz, -7.9dB

8


Output
Debug Output
GEL Output
Serial Console - COM4
Debug Console
Lab 5, Spring 2026
Student EID1= MA69299
Connect PB20 to DACOUT.
Initialization, good, Score=4
Activate piano Key0 Good. Score=10
Activate piano Key1 Good. Score=16
Activate piano Key2 Good. Score=22
Activate piano Key3 Good. Score=25
Done. Score=25

9

measurment data #9

Digital Data	theoretical DAC voltage	measured DAC voltage
0	0V	0V
1	0.106V	0.1V
7	0.745V	0.75V
8	0.852V	0.85V
15	1.597V	1.61V
16	1.703V	1.67V
17	1.810V	1.78V
23	2.448V	2.44V
24	2.555V	2.53V
25	2.661V	2.67V
30	3.194V	3.19V
31	3.30V	3.3V

theory = Data * $\frac{3.3}{31}$

Resolution = $\frac{\text{Range}}{\text{precision}-1}$ = $\frac{3.3}{2^5-1}$ = 0.106V.

(max-min)

Range: 3.3-0 = 3.3V

Precision: 2^5 = 32, 32 distinct output levels.

Accuracy = 100% - 100 $\left(\frac{\text{measured} - \text{theoretical}}{\text{theoretical}}\right)$

= 1.73% < 2%.

so all good.

10

a) When does the interrupt trigger occur here in Lab 5? - The interrupt occurs when the SysTick timer counts down to zero, causing roll over
b) In which file is the interrupt vector? - The SysTick interrupt vector is defined in the startup file (startup_msmpoxxxx.c)
c) List the steps that occur after the trigger occurs and before the processor executes the handler - The processor completes the current instruction. From there, it automatically saves the processor state to the stack, loads the handler address from the vector table, and then branches ot the interrupt service routine
d) how does BX LR return from an interrupt? - BX LR returns from interrupt using a special exception return value in LR that causes the processor to restore that saved context from the stack and resume normal execution.